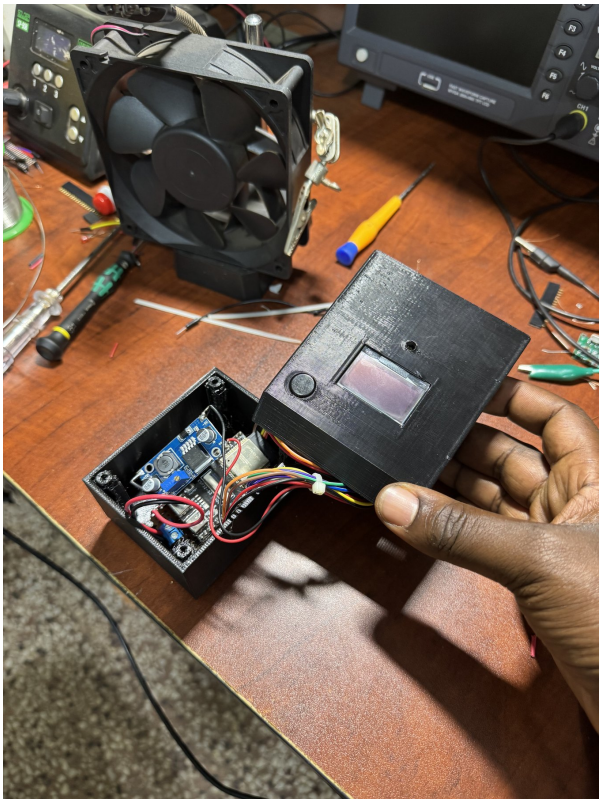
ESP32-S3 XiaoZhi

Illustrated hardware setup for the ESP32-S3-N16R8, INMP441 microphone, MAX98357A amplifier, and SH1106 128x64 OLED



Completed enclosure during final assembly. The display, button, controller, and internal wiring remain accessible for inspection.

CONTROLLER	AUDIO INPUT	AUDIO OUTPUT	DISPLAY
ESP32-S3-N16R8	INMP441 I2S	MAX98357A I2S	SH1106 128x64

This guide documents mpuugaAbdu's hardware customization of the XiaoZhi ESP32 project. Power off the board before moving wires. Confirm module labels because breakout-board pin order and accepted supply voltage can vary.

Signal map

INMP441 digital microphone

Module pin	ESP32-S3	Purpose
WS	GPIO 41	I2S word select / left-right clock
SCK	GPIO 3	I2S serial clock
SD	GPIO 40	Microphone data into the ESP32-S3
VDD	3.3 V	Microphone power
GND	GND	Common ground

The INMP441 L/R pin selects which I2S channel carries samples. If audio is silent, verify that selection against the firmware channel configuration; do not change signal pins at random.

MAX98357A amplifier

Module pin	ESP32-S3	Purpose
DIN	GPIO 42	Digital audio from the ESP32-S3
BCLK	GPIO 1	I2S bit clock
LRC / LRCLK	GPIO 2	I2S left-right clock
GND	GND	Common ground
VIN	Per module label	Use a clean supply within the breakout's rating

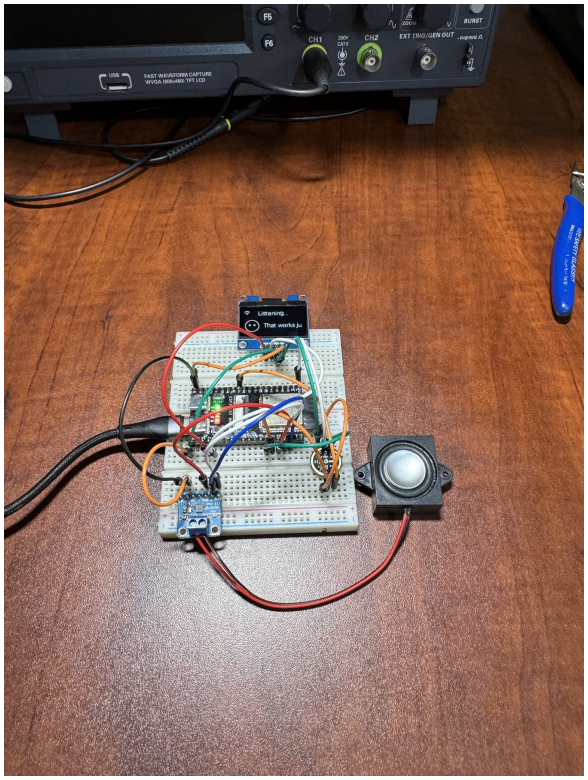
Speaker rule: connect the speaker only between SPK+ and SPK-. The MAX98357A uses a bridge output, so neither speaker terminal should be tied to ground.

SH1106 OLED

Module pin	ESP32-S3	Purpose
SDA	GPIO 21	I2C data
SCL	GPIO 47	I2C clock
VCC	3.3 V	Display power / logic
GND	GND	Common ground

The selected firmware target is SH1106 128x64. Many modules answer at address 0x3C; confirm your module with an I2C scan if the display stays blank.

Build sequence



1. Breadboard validation - prove the display, microphone, amplifier, and speaker independently before soldering.

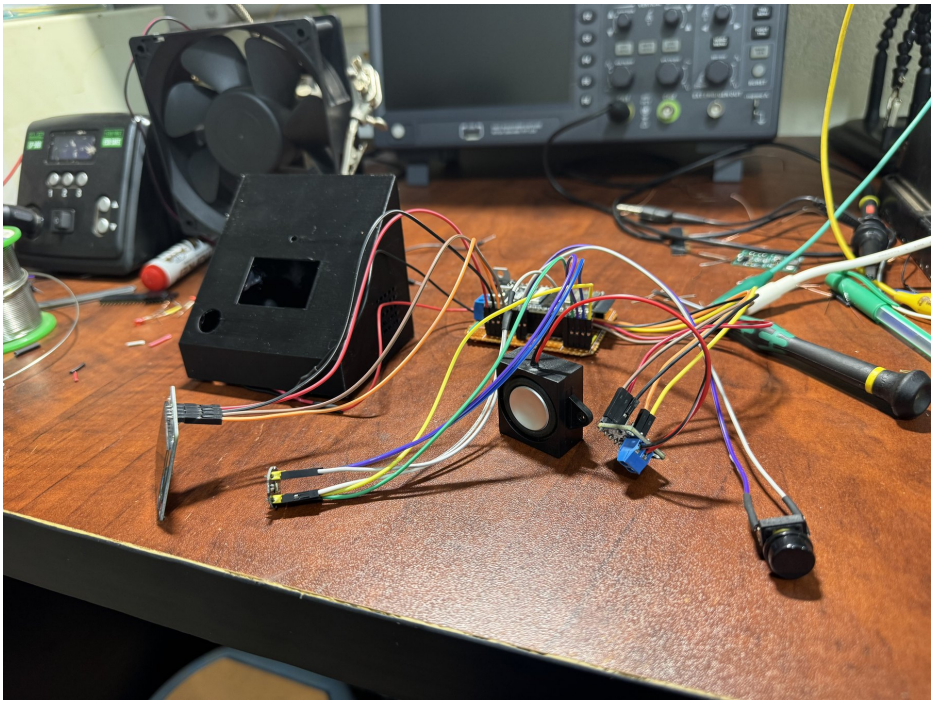
1. Use short I2S clock wires and keep microphone wiring away from the speaker leads.

2. Use one common ground for the ESP32-S3 and all peripheral modules.

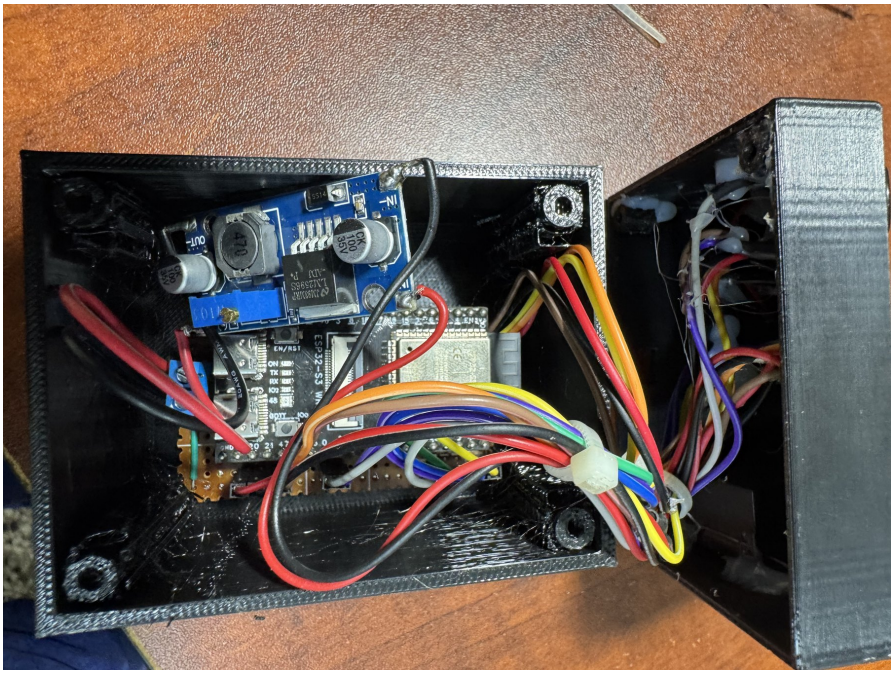
3. Start at low speaker volume to reduce feedback and protect the speaker.

4. Verify that USB power is disconnected before continuity checks or rewiring.

Permanent assembly



2. Transfer the tested circuit to permanent wiring. Label each connector and add strain relief before enclosure installation.



3. Arrange the ESP32-S3, power circuitry, and cable bundle so solder joints cannot contact the enclosure or pull against headers.

Pre-power checklist

Check	Status	Pass condition
Power rails	[ ]	No short; voltage matches every module.
Grounding	[ ]	ESP32-S3 and all peripherals share ground.
Microphone	[ ]	WS=41, SCK=3, SD=40; stable capture.
Amplifier	[ ]	DIN=42, BCLK=1, LRCLK=2; speaker only across SPK+ / SPK-.
OLED	[ ]	SDA=21, SCL=47; SH1106 and address verified.
Mechanical	[ ]	No exposed conductor can touch another signal.

Bring-up order

1. OLED test | 2. Microphone capture | 3. Low-volume playback | 4. Full voice-assistant session

Firmware configuration

The board uses simplex I2S with separate microphone and speaker buses. Build the SH1106 variant with:

```
python3 scripts/build.py bread-compact-wifi --name bread-compact-wifi-sh1106-128x64
```

Attribution

Customized hardware, assembly, and documentation: mpuugaAbdu (github.com/mpuugaAbdu). Original XiaoZhi software copyright (c) 2025 Shenzhen Xinzhi Future Technology Co., Ltd. and Project Contributors. Distributed under the repository's MIT License.